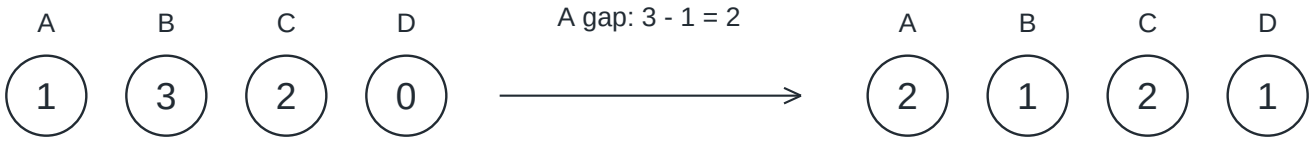
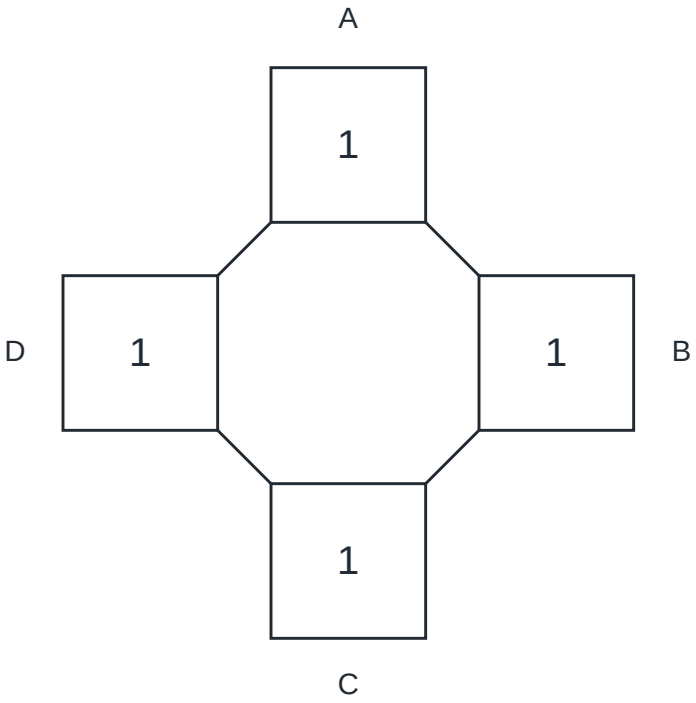
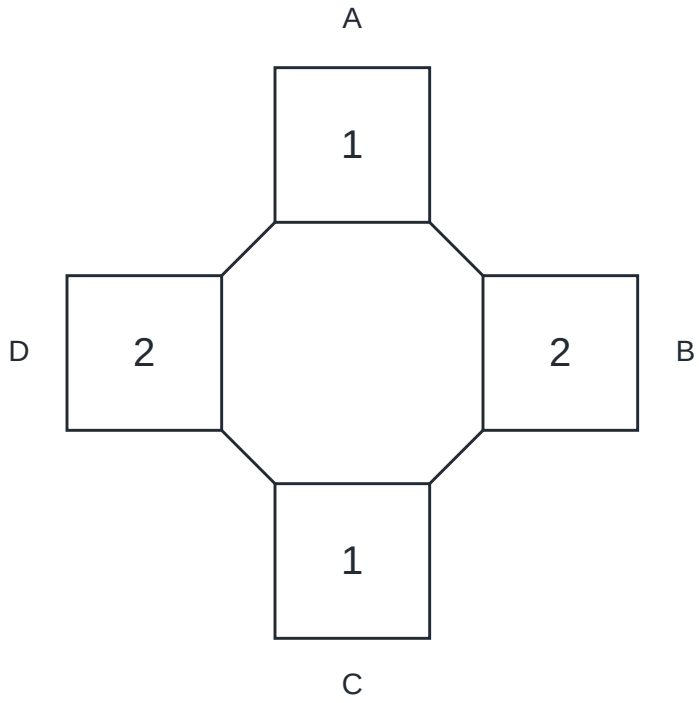
At each station, the new height is the gap between its old height and the next old height, in A-B-C-D-A order. Keep the whole old ring unchanged until every new height is ready.



Problem 1: Find every old ring that makes each printed new ring. Use whole heights from 0 to 3, with at least one 0. Two rings count as different when their A, B, C, D heights differ.



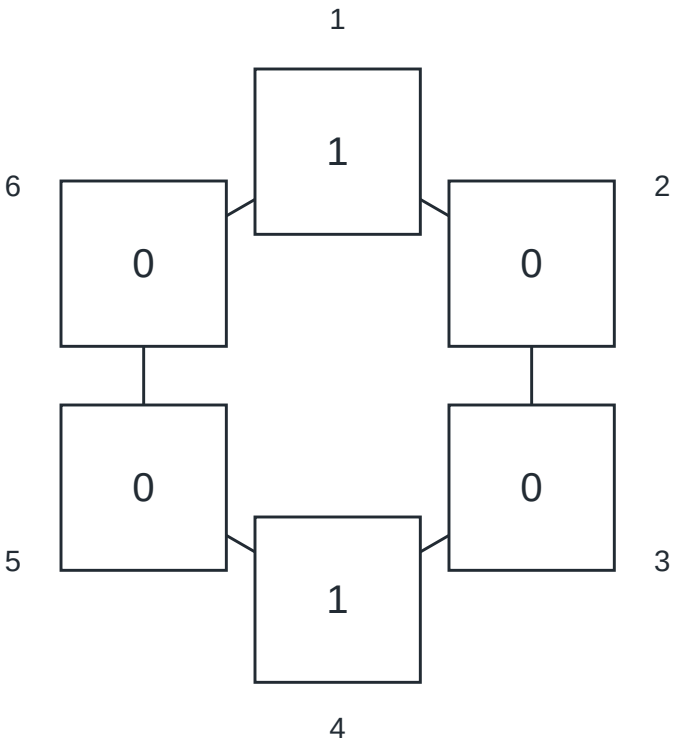


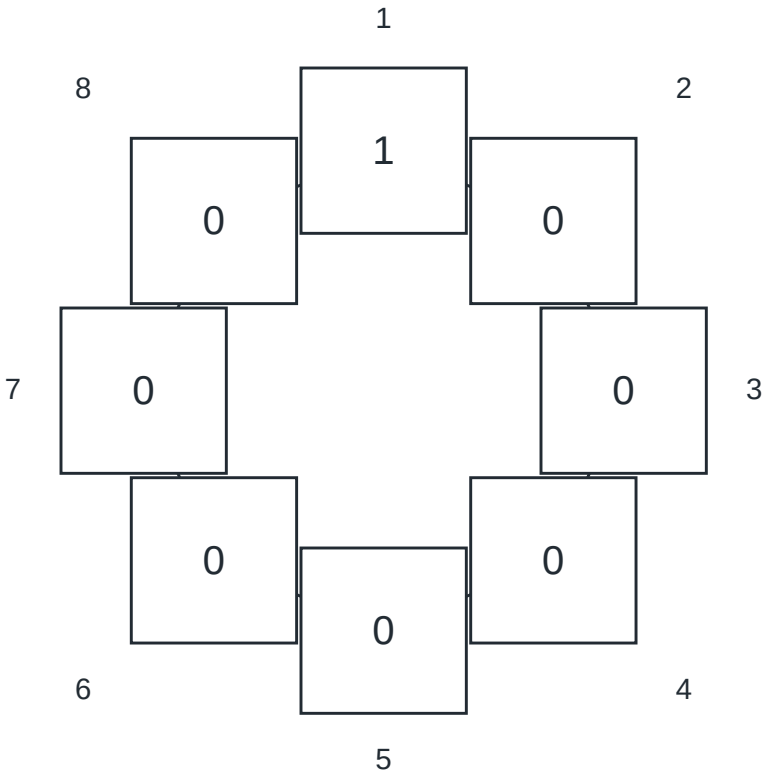
A	B	C	D

A	B	C	D

Use only heights 0 and 1. At each station compare its old height with the next old height in number order, wrapping back to 1. Equal gives 0; different gives 1. Keep the old ring unchanged until every new height is ready.

Problem 2: Try the six-station and eight-station starts. Can a six-station ring repeat without becoming all 0? Can an eight-station start still have a 1 after seven rounds? Decide whether every eight-station start is all 0 by round eight, and explain why.

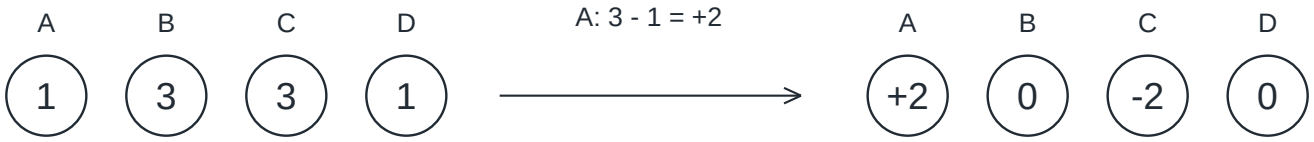




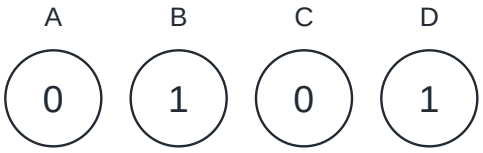
	1	2	3	4	5	6
0						
1						
2						
3						
4						
5						
6						
7						
8						

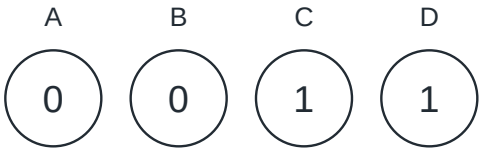
	1	2	3	4	5	6	7	8
0								
1								
2								
3								
4								
5								
6								
7								
8								

Now keep the direction: new = next old - current old. A rise of 2 is +2; a fall of 2 is -2. All four new values still use the same old ring.



Problem 3: Run both starts with the directed rule. Which start gets an entry more than 10 away from 0 by round 5? Can that run get farther from 0 than any chosen ceiling? Explain why your pattern keeps going.





	A	B	C	D
0				
1				
2				
3				
4				
5				
6				
7				

	A	B	C	D
0				
1				
2				
3				
4				
5				
6				
7				